

COMPUTABLE FUNCTIONS

Dr. A. Beulah
AP/CSE

LEARNING OBJECTIVE

- To Design Turing machines for any Languages (K3)
 - To Understand the concept of Turing Machine

TURING-COMPUTABLE FUNCTIONS

- A total function $f: \Sigma^* \rightarrow \Sigma^*$ is Turing-computable if there exists a DTM M such that for every x in Σ^* ,

$$(\underline{s}, B\underline{x}B) \vdash^* (\underline{h}, B\underline{f(x)}B).$$

$$L = \{0^n 1^n \mid n \geq 1\}$$
$$L = \{1^n 2^n 3^n\} \quad \Delta, \Sigma$$

- A partial $f: \Omega \rightarrow \Sigma^*$ is Turing-computable if there exists a DTM M such that $L(M) = \Omega$ and for every x in Ω ,

$$(\underline{s}, B\underline{x}B) \vdash^* (\underline{h}, B\underline{f(x)}B).$$

TURING-COMPUTABLE FUNCTIONS

Construct a TM for successive function ?

$$f : \underline{\mathbb{N}} \rightarrow \underline{\mathbb{N}}, f(\underline{x}) = \underline{x+1}.$$

Assume that the input is encoded in UNARY form.

Let $M = (\underline{Q}, \underline{\Sigma}, \Gamma, \delta, q_0, B, F)$

$Q = \{q_0, q_1\}$ q_0 =start state q_1 =final state

$\Sigma = \{0\}$

$\Gamma = \{0, B\}$

$F = \{q_1\}$

$$\begin{aligned} \delta(q_0, 0) &= (q_0, 0, R) \\ \delta(q_0, B) &= (q_1, 0, R) \end{aligned}$$

$$\begin{aligned} \frac{2}{2} & \quad \frac{f(2)}{2+1} \\ 5 & \rightarrow 6 \\ 2 & \rightarrow 3 \end{aligned}$$

$$\underline{4 \rightarrow 5}$$

0, 1, 2, 3, ... 9

1
2
3
4

0
00
000
0000

0 0 0 0 0

5

2 2+1
00 000
000
000

TURING-COMPUTABLE FUNCTIONS

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Handwritten notes illustrating the unary representation of 2 and its successor 3:

00 2, 2+1
 $q_0 \underline{00} \vdash 0 q_0 \underline{0}$
 $\vdash 00 q_0 B$
 $\vdash \underline{000} q_1 B$

States	Tape Symbols	
	<u>0</u>	<u>B</u>
q_0	$(q_0, \underline{0}, R)$	$(\underline{q_1}, \underline{0}, R)$
q_1	—	—

TURING-COMPUTABLE FUNCTIONS

Let us consider the input $x=\underline{3}$, This is encoded as 000.

$$\begin{aligned}(q_0, \underline{000}B) &\vdash (q_0, 0\underline{00}B) \vdash (q_0, 00\underline{0}B) \\ &\vdash (q_0, 000\underline{B}) \vdash (\underline{q_1}, \underline{0000}B)\end{aligned}$$

The machine halts in an accepting state q_1
by computing the successive of x

ADDITION

- We can represent numbers on a TM by using the unary representation, which just uses n 0's to represent the number n .

Example: 00000 = 5

- We can construct a TM to concatenate two strings together.
- If we represent two numbers in unary form, then a TM which concatenates these two strings is actually performing addition!

$\rightarrow m, n$
 $f(m, n) = \underline{m+n}$

$3+2$
m n
000100
000100
00000

$3+0$
00100 $\rightarrow$ 0000
 $2+2$
00100 $\rightarrow$ 0000

ADDITION

- Addition of two numbers as simple as concatenation of two strings.
- Assume two integers are represented in unary form on the tape of a Turing machine.
- Assume the two integers are separated by a 1 between them.
- Design a Turing machine that will add these two numbers.

ADDITION

- $f(m,n)=m+n$

$$in \quad \cdot O' \rightarrow \beta$$

$$\delta(q_0, 0) = (q_1, B, R)$$

$$\delta(a_1, 0) = (a_1, 0, R)$$

$$\delta(q_1, \perp) = (q_2, 0, R)$$

$$B(q_1, 0) = (q_2, 0, R)$$

$$\delta(a_2, B) = (a_3, B, R)$$

000100
 000000

 000100
 $\downarrow$
 $B00000$
 $\rightarrow F$
 000

0010
 $q_0 0010$
 $\vdash B q_1 010$
 $\vdash B 0 q_1 10$
 $\vdash B 00 q_2 0$
 $\vdash B 000 q_2 B$
 $\vdash B 0000 q_3$

$2+1$
 $q_3 \in F$

ADDITION

- 3+2

SUBTRACTION M - N

- For example, proper subtraction $m - n$ is defined to be

$$f(m,n) = \begin{cases} m - n & \text{for } m \geq n, \text{ and } \underline{\underline{00\dots}} \\ \text{zero} & \text{for } m < n. \end{cases}$$

B 0 3

- The TM $M = (\{q_0, q_1, \dots, q_6\}, \{0, 1\}, \{0, 1, B\}, \partial, q_0, B, \{ \})$

3-1

$$\begin{array}{r} 00010 \\ \hline B \quad \uparrow \\ \hline 00 \end{array}$$

The function ∂ is described below.

$\partial(q_0, 0) = (q_1, \underline{B}, R)$ Begin. Replace the leading 0 by B.

$\partial(q_1, 0) = (q_1, 0, R)$ Search right looking for the first 1.

$\partial(q_1, 1) = (\underline{q_2}, 1, R)$

$\partial(q_2, 1) = (q_2, 1, R)$ Search right past 1's until encountering a 0. Change that 0 to 1.

$\partial(q_2, 0) = (\underline{q_3}, \underline{1}, L)$

$\partial(q_3, 0) = (q_3, 0, L)$ Move left to a blank. Enter state q_0 to repeat the cycle.

$\partial(q_3, 1) = (q_3, 1, L)$

$\partial(q_3, B) = (q_0, B, R)$

If in state q_2 a B is encountered before a 0, we have situation i described above. Enter state q_4 and move left, changing all 1's to B's until encountering a B. This B is changed back to a 0, state q_6 is entered and M halts.

$\partial(q_2, B) = (\underline{q_4}, \underline{B}, L)$

$\partial(q_4, 1) = (q_4, B, L)$

$\partial(q_4, 0) = (q_4, 0, L)$

$\partial(q_4, B) = (\underline{q_6}, 0, R)$

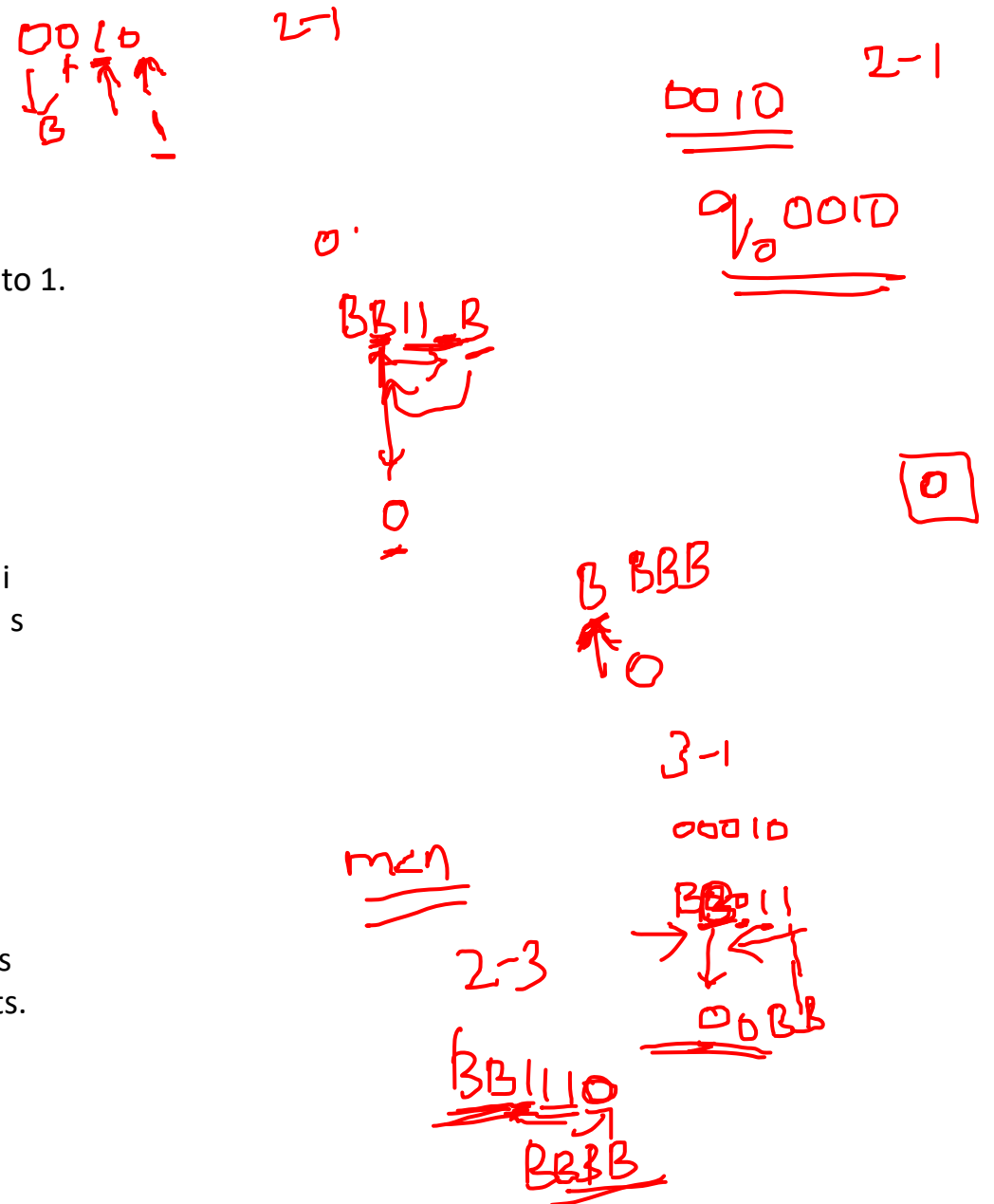
If in state q_0 a 1 is encountered instead of a 0, the first block of 0's has been exhausted, as in situation (ii) above. M enters state q_5 to erase the rest of the tape, then enters q_6 and halts.

$\partial(q_0, 1) = (\underline{q_5}, B, R)$

$\partial(q_5, 0) = (q_5, B, R)$

$\partial(q_5, 1) = (q_5, B, R)$

$\partial(q_5, B) = (\underline{q_6}, B, R)$

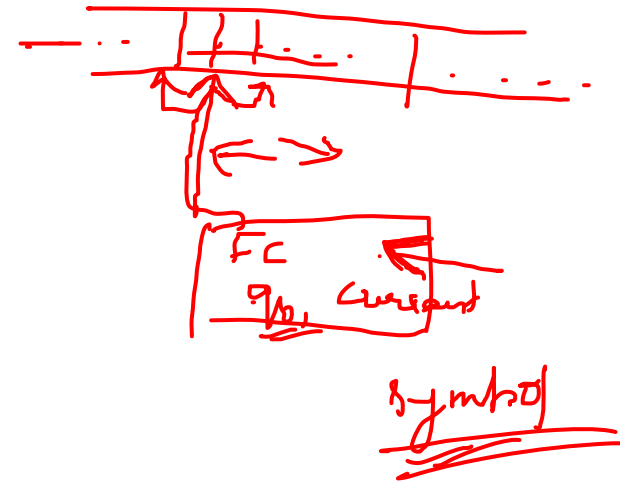


SUBTRACTION M - N

	symbol		
state	0	1	<i>B</i>
q_0	(q_1, B, R)	(q_5, B, R)	-
q_1	$(q_1, 0, R)$	$(q_2, 1, R)$	-
q_2	$(q_3, \textcolor{red}{1}, L)$	$(q_2, 1, R)$	(q_4, B, L)
q_3	$(q_3, 0, L)$	$(q_3, 1, L)$	(q_0, B, R)
q_4	$(q_4, 0, L)$	(q_4, B, L)	$(q_6, 0, R)$
q_5	(q_5, B, R)	(q_5, B, R)	(q_6, B, R)
q_6	-	-	-

PROGRAMMING TECHNIQUES OF TURING MACHINES

- Storage in the Finite Control
- Multiple Tracks
- Checking off Symbols
- Subroutines



STORAGE IN THE FINITE CONTROL

- The finite control can be used to hold the finite amount of information.
- It is considered as a pair of elements, like (q_0, a) , where one exercising *control* and second component stores a symbol in the finite control.
- Consider a turing machine M which accepts the language $01^* + 10^*$
- Let $M = (Q, \{0,1\}, \{0,1,B\}, \delta, \{q_0, B\}, B, F)$
- $Q = \{q_0, q_1\} \times \{0,1,B\}$
 $= ([q_0,0], [q_0,1], [q_0,B], [q_1,0], [q_1,1], [q_1,B])$
- $F = \{[q_1,B]\}$

$[q_0, \text{symbol}]$
 q_0

0.111111
 1000000

$L = \{0, 1, 01, 011, 0111, \dots, 10, 100, 1000, \dots\}$

STORAGE IN THE FINITE CONTROL

$$\delta(\underline{[q_0, B]}, \underline{0}) = (\underline{[q_1, 0]}, \underline{0, R})$$

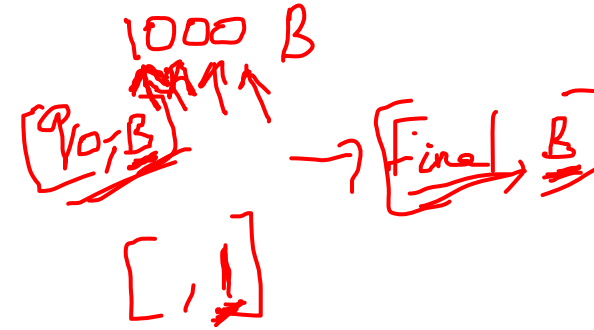
$$\delta(\underline{[q_0, B]}, \underline{1}) = (\underline{[q_1, 1]}, \underline{1, R})$$

$$\delta(\underline{[q_1, 0]}, \underline{1}) = (\underline{[q_1, 0]}, \underline{1, R}) \rightarrow$$

$$\delta(\underline{[q_1, 1]}, \underline{0}) = (\underline{[q_1, 1]}, \underline{0, R}) -$$

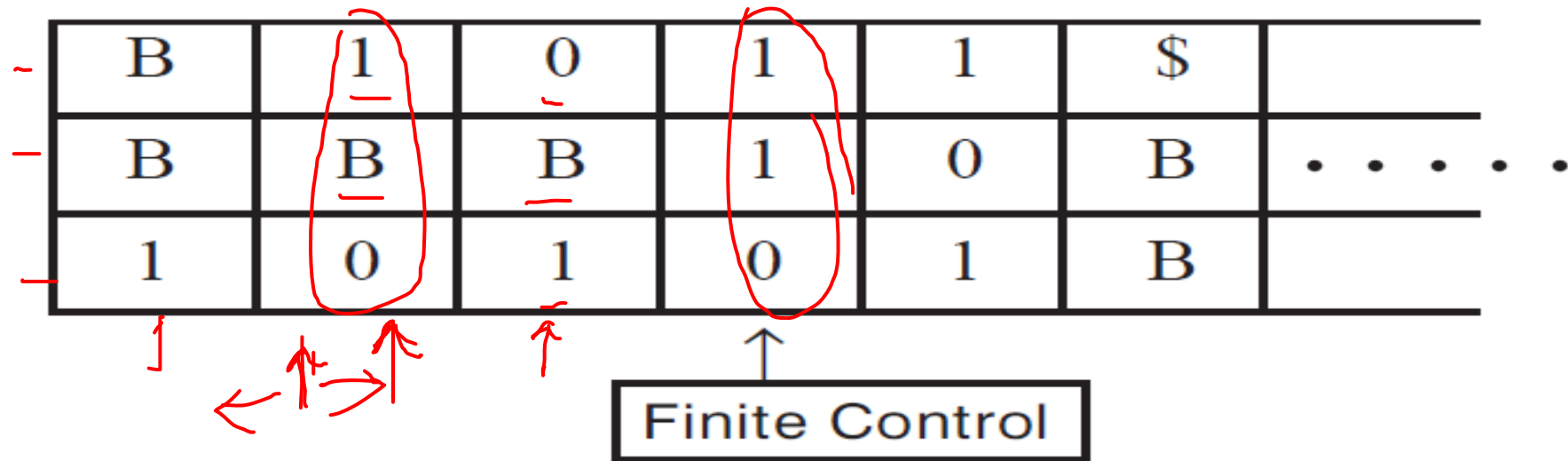
$$\delta(\underline{[q_1, 0]}, \underline{B}) = (\underline{[q_1, B]}, \underline{0, L})$$

$$\delta(\underline{[q_1, 1]}, \underline{B}) = (\underline{[q_1, B]}, \underline{1, L})$$



$$\delta \left(\underline{[q_1, sy]}, \underline{[q_y, sy^2]} \right)$$

MULTIPLE TRACKS



$$\delta(q_0, \underline{0}) = (q_1, \underline{x}, r)$$

$$[B, B, 1]$$

$$\delta(q_0, \underline{[1, 1, 0]})$$

CHECKING OFF SYMBOLS

$\{wcw^R / w \text{ in } \Sigma^*\}$

$L = \{aaa, a, b, aab, ab, \dots\} \in \mathcal{C}(b)$

$\{a^i b^i : i \geq 1\}$

Multiple track

storage in finite control.

$aabbb$
 $x \rightarrow y$
 $x \rightarrow y$

B	B	B	B	✓	✓	
a	a	b	c	a	b	

extra Track

✓ ✓ ✓ ✓
~~B~~ ~~B~~ ~~B~~ ~~B~~ B
a a b b B
 $x a$
 $\underline{B B}$
term

CHECKING OFF SYMBOLS

Consider a turing machine $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$ for the language

$$L = \{\underline{wcw} \mid w \in \{a, b\}^+\}$$

a) $Q = \{[q, d] \mid q = q_1 q_2 \dots q_9 \text{ and } d = a, b \text{ or } B\}$

b) $\Sigma = \{[B, d] \mid d = a, b \text{ or } c\}$

c) $\Gamma = \{[x, d] \mid x = B \text{ or } \text{ and } d = a, b, c \text{ or } B\}$

d) $q_0 = [q_1, B]$

e) $F = \{[q_9, B]\}$

f) $B = [B, B]$

g) δ is defined for $d = a \text{ or } b$ and $e = a \text{ or } b$.

CHECKING OFF SYMBOLS

Forward

$d \rightarrow e / b$

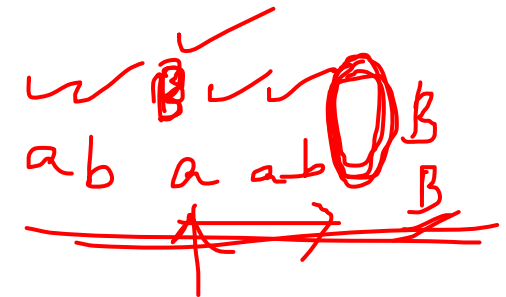
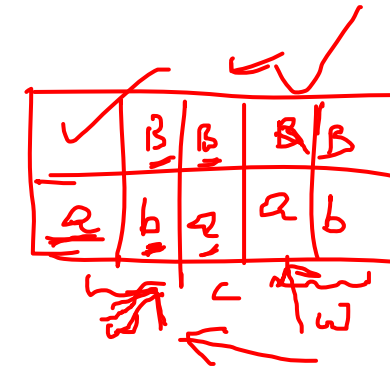
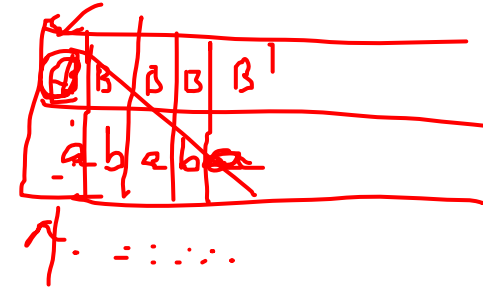
$$\delta([q_1, \underline{B}], [\underline{B}, \underline{d}]) = ([q_2, \underline{d}], [\underline{\checkmark}, \underline{d}], R)$$

$$\delta([q_2, d], [B, \underline{e}]) = ([q_2, d], [B, e], R)$$

$$\delta([q_2, d], [B, \underline{c}]) = ([q_3, d], [B, c], R)$$

$$\delta([q_3, d], [\underline{\checkmark}, e]) = ([q_3, d], [\underline{\checkmark}, e], R)$$

$$\delta([q_3, d], [\underline{B}, \underline{d}]) = ([q_4, B], [\checkmark, d], \underline{L})$$



CHECKING OFF SYMBOLS

Backward

$$\delta([q_4, B], [\checkmark, d]) = ([q_4, B], [\checkmark, d], L)$$

$$\delta([q_4, B], [B, c]) = ([q_5, B], [B, c], L)$$

$$\delta([q_5, B], [B, d]) = ([q_6, B], [B, d], L)$$

$$\delta([q_6, B], [B, d]) = ([q_6, B], [B, d], L)$$

$$\delta([q_6, B], [\checkmark, d]) = ([\underline{q_1}, B], [\checkmark, d], R)$$

CHECKING OFF SYMBOLS

Terminate

$$\delta([q_5, B], [\checkmark, d]) = ([q_7, B], [\checkmark, d], R)$$

$$\delta([q_7, B], [B, c]) = ([q_8, B], [B, c], R)$$

$$\delta([q_8, B], [\checkmark, d]) = ([q_8, B], [\checkmark, d], R)$$

$$\delta([q_8, B], [\underline{B}, \underline{B}]) = ([\underline{q_9}, \underline{B}], [, B], L)$$

SUBROUTINES

$\delta(q0,0) = (q6, B, R)$

$\delta(q6, 0) = (q6, 0, R)$

$\delta(q6, 1) = (q1, 1, R)$

$f(m,n) = m+n$

$f(m,n) = m \times n$

$$\begin{array}{r} \overline{m} \mid n \\ \hline \overline{000} \mid \overline{00} \\ \hline \end{array}$$

$$\begin{array}{r} \overline{m \times n} \\ \hline \overline{000000} \end{array}$$

$2 \times 2 = 4$
 $m \quad n$
 001001 result
 2×2
 $m \times n$
 00100100
 $m \times n$
 0010010000
 0000

δ for subroutine COPY.

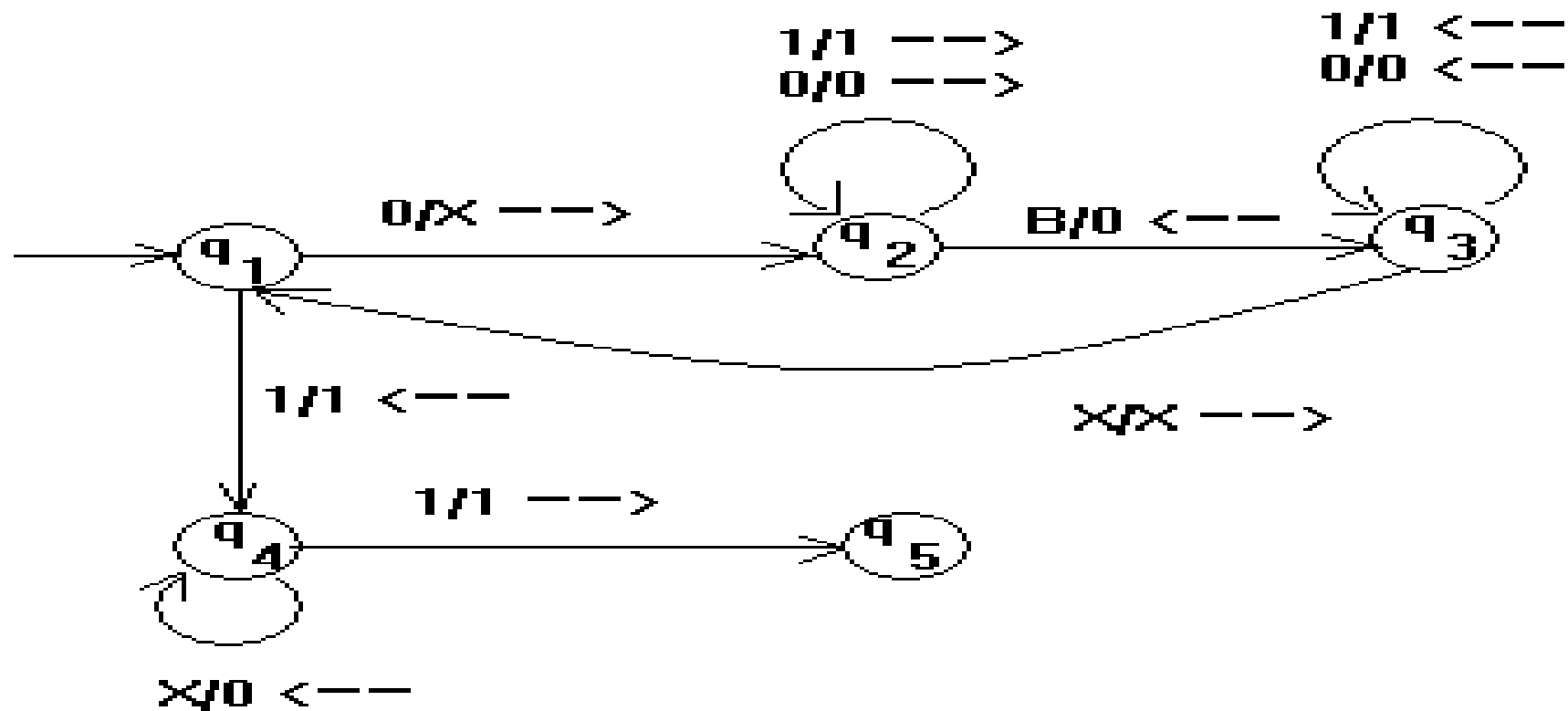
States	Inputs			
	0	1	2	B
q_1	$(q_2, 2, R)$	$(q_4, 1, L)$		
q_2	$(q_2, 0, R)$	$(q_2, 1, R)$		$(q_3, 0, L)$
q_3	$(q_3, 0, L)$	$(q_3, 1, L)$	$(q_1, 2, R)$	
q_4		$(q_5, 1, R)$	$(q_4, 0, L)$	



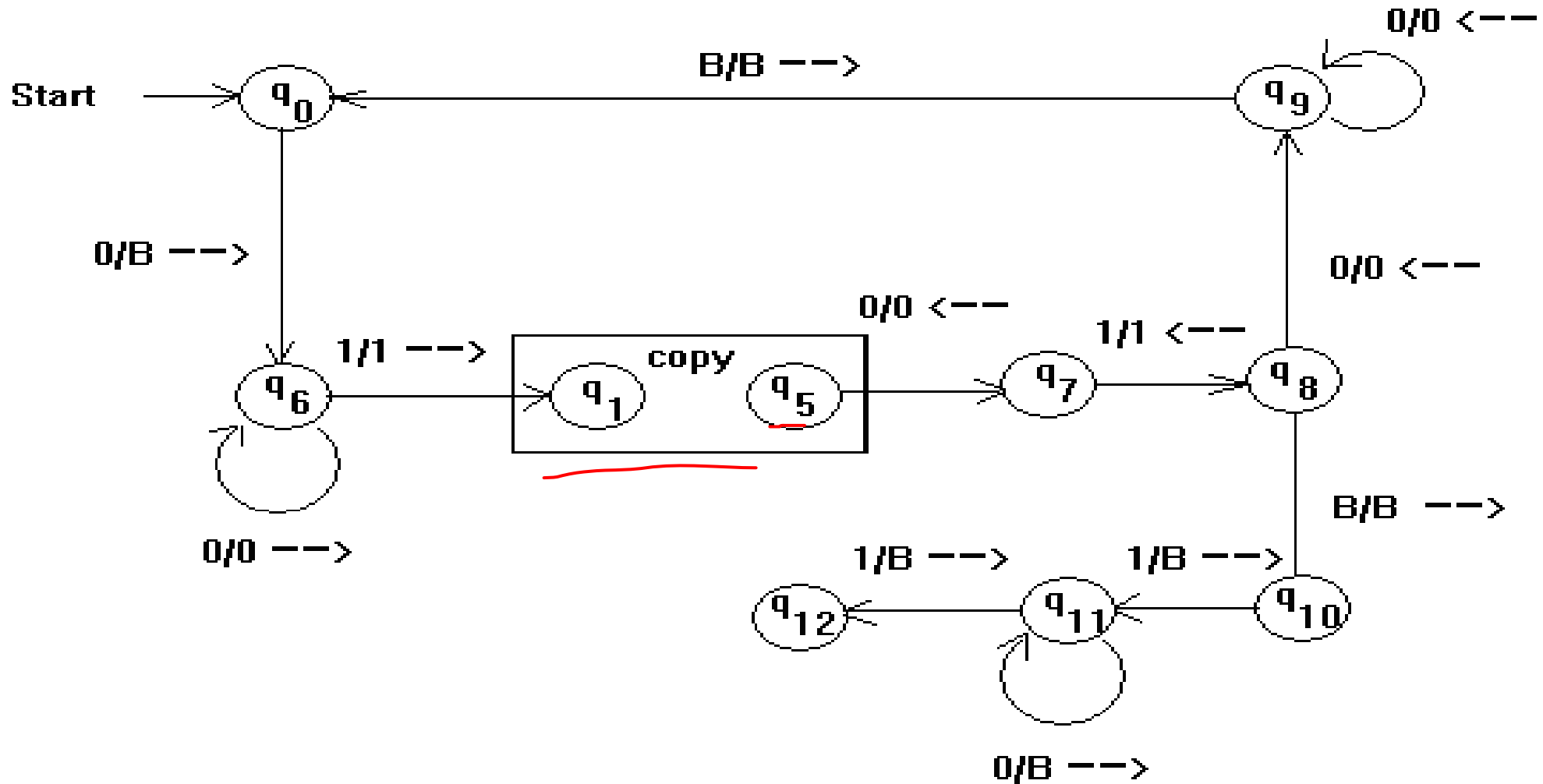
SUBROUTINES

States	Inputs			
	0	1	2	B
q_5	$(q_7, 0, L)$			
q_7		$(q_8, 1, L)$		
q_8	$(q_9, 0, L)$			(q_{10}, B, R)
q_9	$(q_9, 0, L)$			(q_0, B, R)
q_{10}		(q_{11}, B, R)		
q_{11}	(q_{11}, B, R)	(q_{12}, B, R)		

SUBROUTINES



SUBROUTINES



$\delta(q_0, \underline{001001}) \vdash Bq_6 01001B$

$\vdash B0q_6 \underline{1001B}$
 $\vdash B01q_1 \underline{001B}$
 $\vdash B01Xq_2 01B$
 $\vdash B01X0q_2 1B$
 $\vdash B01X01q_2 B$
 $\vdash B01X0q_3 10$
 $\vdash B01Xq_3 010$
 $\vdash B01q_3 X010$
 $\vdash B01Xq_1 010$
 $\vdash B01XXq_2 10$
 $\vdash B01XX1q_2 0$
 $\vdash B01XX10q_2 B$
 $\vdash B01XX1q_3 \underline{00}$
 $\vdash B01XXq_3 100$
 $\vdash B01Xq_3 X100$
 $\vdash B01XXq_1 100$
 $\vdash B01Xq_4 X100$
 $\vdash B01q_4 X0100$
 $\vdash \underline{B0q_4 100100}$
 $\vdash B01q_5 00100$
 $\vdash B0q_7 100100$

2x2

$\vdash Bq_8 \underline{0100100}$
 $\vdash q_9 B0100100 ($
 $\vdash Bq_0 0100100$
 $\vdash BBq_6 \underline{100100}$
 $\vdash BB1q_1 00100$
 $\vdash BB1Xq_2 0100$
 $\vdash BB1X0q_2 100$
 $\vdash BB1X0q_2 100$
 $\vdash BB1X01q_2 00$
 $\vdash BB1X010q_2 0$
 $\vdash BB1X0100q_2 B$
 $\vdash BB1X010q_3 00$
 $\vdash BB1X01q_3 000$
 $\vdash BB1X0q_3 1000$
 $\vdash BB1Xq_3 01000$
 $\vdash BB1q_3 X01000$
 $\vdash BB1Xq_1 01000$
 $\vdash BB1XXq_2 1000$
 $\vdash BB1XX1q_2 000$
 $\vdash BB1XX10q_2 00$
 $\vdash BB1XX100q_2 0$
 $\vdash BB1XX1000q_2 B$
 $\vdash BB1XX100q_3 00$
 $\vdash BB1XX10q_3 000$
 $\vdash BB1XX1q_3 0000$
 $\vdash BB1XXq_3 10000$
 $\vdash BB1Xq_3 X10000$
 $\vdash BB1XXq_1 10000$

SUBROUTINES

- | – $BB1Xq_4X10000$
- | – $BB1q_4X010000$
- | – $BBq_410010000$
- | – $BB1q_50010000$
- | – $BBq_710010000$
- | – $Bq_8B10010000$
- | – $BBq_{10}10010000$
- | – $BBBq_{11}0010000$
- | – $BBBBq_{11}010000$
- | – $BBBBBq_{11}10000$
- | – $BBBBBBq_{12}0000$

SUMMARY

- Definition of Total and partial function
- Computing a numerical function using Turing Machine
- Programming techniques of Turing Machine

TEST YOUR KNOWLEDGE

- Which of the following statements is/are FALSE?
 1. For every non-deterministic Turing machine, there exists an equivalent deterministic Turing machine.
 2. Turing recognizable languages are closed under union and complementation.
 3. Turing decidable languages are closed under intersection and complementation.
 4. Turing recognizable languages are closed under union and intersection.
- A. 1 and 4 only
B. 1 and 3 only
C. 2 only
D. 3 only

TEST YOUR KNOWLEDGE

- Which of the following is true for the language
 $L = \{a^p / p \text{ is prime}\}$
 - A. It is not accepted by a Turing Machine
 - B. It is regular but not context-free
 - C. It is context-free but not regular
 - D. It is neither regular nor context-free, but accepted by a Turing machine

REFERENCE

- Hopcroft J.E., Motwani R. and Ullman J.D, “Introduction to Automata Theory, Languages and Computations”, Second Edition, Pearson Education, 2008